

Hysteresis Lab Report

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Due Date: May 2, 2026, 11:59 PM

1 Introduction

On Friday, May 1, 2026 an experiment was conducted with the goal of observing and measuring the hysteresis loops, magnetisation curves, and Curie temperatures of various unknown ferrite materials.

Ferromagnetism is a property of some materials that results in significant magnetic permeability and coercivity [1], which is the amount of induced magnetism in a material in response to an external magnetic field [2] and its ability to resist becoming demagnetised [3], respectively, allowing it to form a permanent magnet. It is the phenomenon most associated with *magnetism* in the vernacular. The Curie temperature of a ferromagnetic material is the point at which the thermal interactions of the atoms overwhelm the materials coercivity, resulting in it losing its magnetic properties [4].

A hysteresis loop is a plot of external magnetic field (H) vs. magnetisation (B) [5]. (fig. 1)

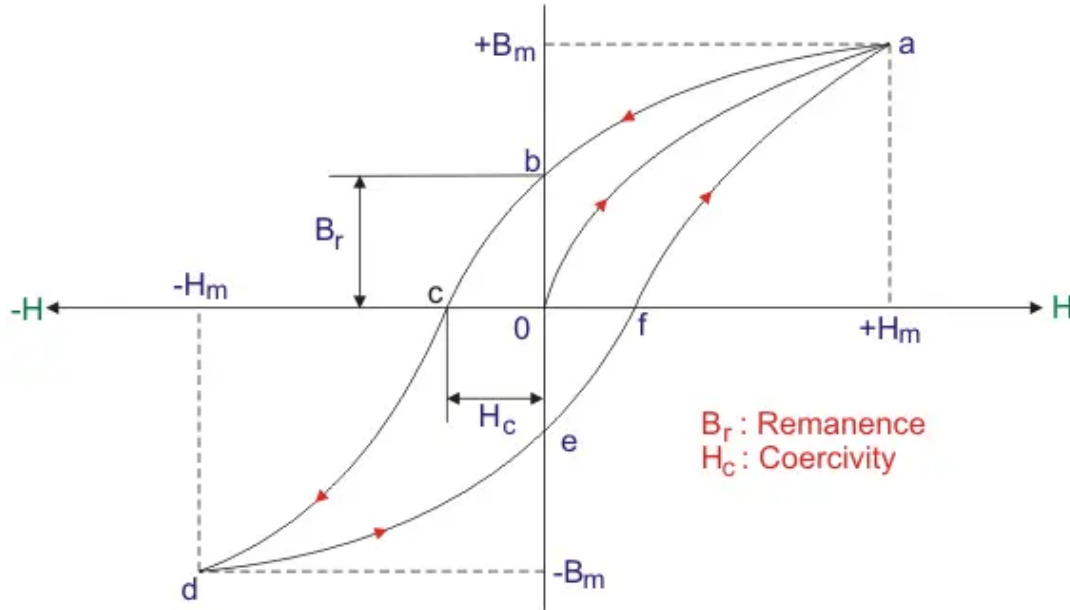


Figure 1: Example Hysteresis Curve.

The magnetisation associated with 0 external magnetic field is called remanence (b). This is the magnetisation of a permanent magnet. The point at which no more magnetisation can be induced regardless of the strength of the external field is called saturation (a). The coercivity, or coercive force, is $|c|$, which is the opposite field strength required to demagnetise the ferromagnet.

The magnetisation curve is the path traced by the saturation point (a) when the external magnetic field strength is increased. It is represented by the line from the origin to a in the example. [5] [4]

1.1 Equipment List

1. Digital Oscilloscope
2. LEAI-71 (Signal Generator (a) and Hysteresis Loop Measurement Circuit (b)) (fig. 2)
3. Assorted cables



Figure 2: LEAI-71 before any wiring or connections. [4]

2 Method and Results

2.1 Method for Hysteresis Loop and Magnetisation Curve Measurement

1. The LEAI-71 was wired according to Figure 10 in the Lab Manual. The oscilloscope was set up according to the steps laid out in the lab manual. The LEAI-71 was switched on to warm up for 10 minutes.
2. The sample was connected to the LEAI-71 per Figure 10.
3. The frequency was set to 7514 Hz (lowest possible) and the magnetising current set to 0.
4. The channel 1 and channel 2 grid spacings were set to 20mV and 10mV respectively. Channel 1 was adjusted -5.25mV and channel 2 -300mV to place the data signal at the origin.
5. Magnetising current was increased to maximum and grid spacing adjusted to 60mV/12.5mV to fit the entire curve to the screen.
6. The sample was demagnetised by slowly reducing the magnetising current.
7. The magnetising current was slowly increased until saturation.
8. Oscilloscope cursors were used to measure apparent H_m, B_m, H_c, B_r

9. The sample was demagnetised again.
10. The magnetising current was slowly increased until saturation, with readings of apparent H_m, B_m every 5mV to measure magnetising curve.
11. Sample was demagnetised and swapped. Steps 2-11 were repeated for Samples A and B.

2.2 Hysteresis Loop Results

Sample	H_m (mV)	B_m (mV)	H_c (mV)	B_r (mV)
A	318.75	45.775	-54.000	27.175
B	369.90	47.025	-29.400	24.675

Table 1: Apparent (measured) hysteresis loop results for Sample A and B.

These apparent values were converted by the given equations

$$H = \frac{N_1 \cdot V_1}{l \cdot R_1} \quad (1)$$

$$B = \frac{R_2 \cdot C \cdot V_2}{N_2 \cdot S} \quad (2)$$

Where $l = 13.8\text{mm}$, $S = 6.4\text{mm}^2$, $C = 1\mu F$, $R_1 = 2\Omega$, $R_2 = 300\Omega$, $N_1 = N_2 = 10$ and V_1 is the x -component measured from the oscilloscope (H_m, H_c) and V_2 is the y -component (B_m, B_r).

These carry dimensions

$$H = \frac{\text{mV}}{\text{mm} \cdot \Omega} = \frac{\text{Amperes}}{\text{Meter}}$$

$$B = \frac{\Omega \cdot \mu F \cdot \text{mV}}{\text{mm}^2} = \text{m} \frac{\text{Webber}}{\text{Meter}^2} = \text{mTesla}$$

Giving the converted results

Sample	$H_{\text{saturation}}$ (Am^{-1})	$B_{\text{saturation}}$ (mT)	$H_{\text{coercivity}}$ (Am^{-1})	$B_{\text{remanance}}$ (mT)
A	115.49	214.57	-19.565	127.38
B	134.02	220.43	-10.652	115.66

Table 2: Converted hysteresis loops results (via (1) and (2)) for Samples A and B.

The raw data was saved from the oscilloscope and parsed with a python script available at https://codeberg.org/londonwallace/bmath/src/branch/main/PHYS2001/Hysteresis/convert_csv.py and plotted. The B_m, H_m, B_r , and H_c points were marked using the measured value in the table above and were not re-derived from the raw CSV data.

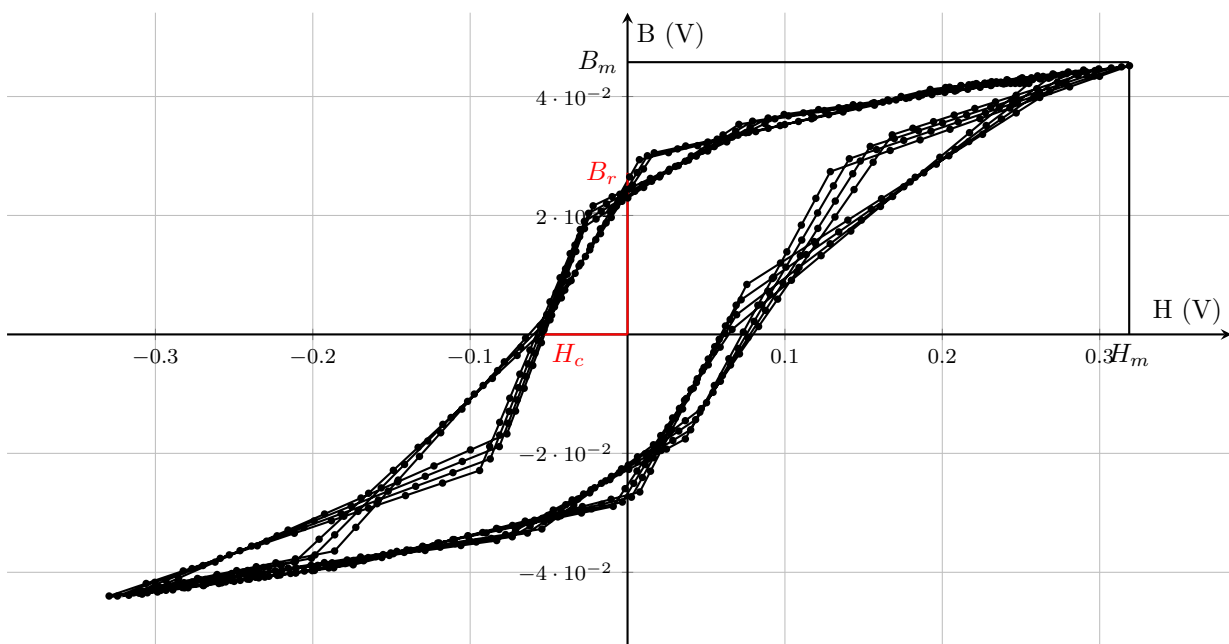


Figure 3: Sample A Hysteresis Plot.

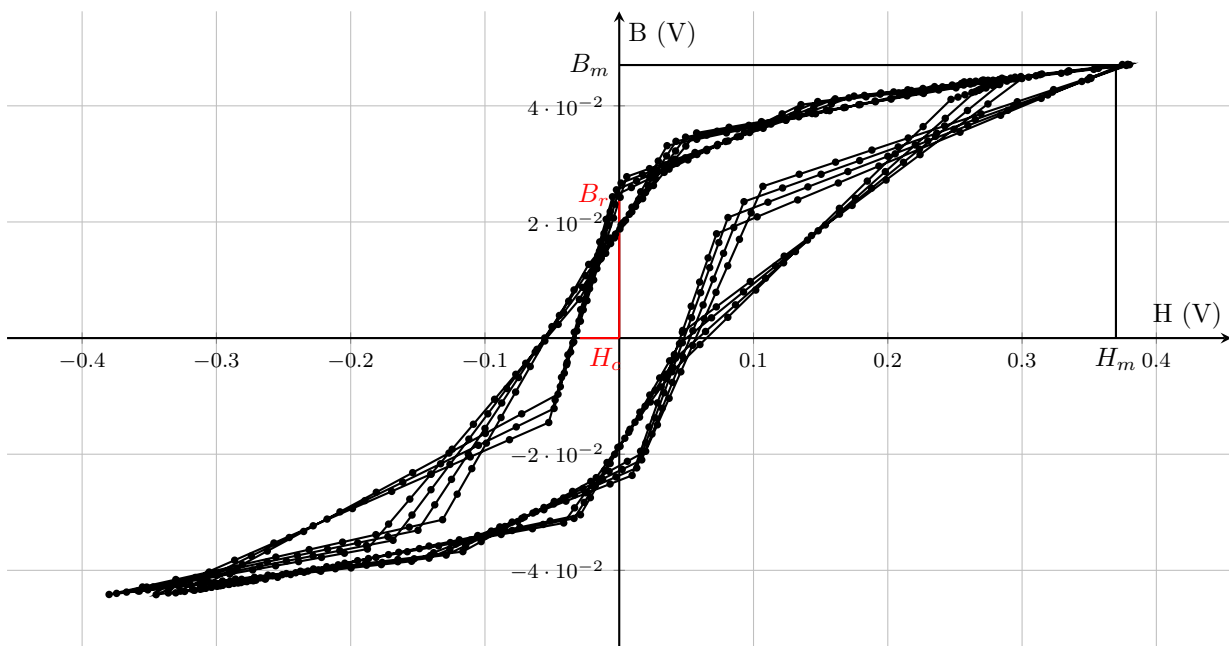


Figure 4: Sample B Hysteresis Plot.

2.3 Magnetising Curve Results

Sample A		Sample B	
H_m (mV)	B_m (mV)	H_m (mV)	B_m (mV)
5	28	5	22
10	48	10	34
15	63	20	50
20	75	25	57
25	87	30	64
30	96	35	71
35	106	50	79
40	117	45	86
45	129	50	94
50	140	55	104
55	154	60	118
60	171	65	135
65	196	70	156
70	230	75	194
75	285	80	265
80	370	85	390
85	504	90	654
90	745	91	766

Table 3: Apparent (measured) magnetising curve results for Sample A (left) and Sample B (right).

Equations (1) and (2) were used, as well as

$$\mu = \frac{B}{H} \quad (3)$$

To convert these apparent values to their respective H, B, and μ . Permeability μ carries the dimensions

$$\frac{\frac{\text{mT}}{\frac{\text{Amperes}}{\text{Meter}}}}{\text{Meter}} = \frac{\text{mT} \cdot \text{m}}{\text{A}} = \frac{\text{mH}}{\text{m}}$$

Results of this conversion are found in table 4. These converted values of magnetisation and permeability were plotted against external field strength for Sample A (fig. 5) and Sample B (fig. 6). Permeability was scaled by a factor of 10 to better illustrate the relationship between permeability and magnetisation on the available grid spacing.

Sample A			Sample B		
H_m (Am ⁻¹)	B_m (mT)	μ (mH·m ⁻¹)	H_m (Am ⁻¹)	B_m (mT)	μ (mH·m ⁻¹)
1.81	28	15.46	1.81	22	12.15
3.62	48	13.25	3.62	34	9.39
5.43	63	11.60	7.24	50	6.90
7.24	75	10.35	9.05	57	6.29
9.05	87	9.61	10.86	64	5.89
10.86	96	8.83	12.67	71	5.60
12.67	106	8.36	14.48	79	5.45
14.48	117	8.08	16.29	86	5.27
16.29	129	7.91	18.1	94	5.19
18.1	140	7.73	19.91	104	5.22
19.91	154	7.73	21.72	118	5.43
21.72	171	7.87	23.52	135	5.737
23.52	196	8.32	25.34	156	6.15
25.34	230	9.07	27.15	194	7.14
27.15	285	10.49	28.96	265	9.15
28.96	370	12.77	30.77	390	12.67
30.77	504	16.37	32.58	654	20.07
32.58	745	22.86	32.94	766	23.25

Table 4: Converted magnetic field strength (H) and magnetisation (B), and derived permeability (μ) for Sample A (left) and Sample B (right).

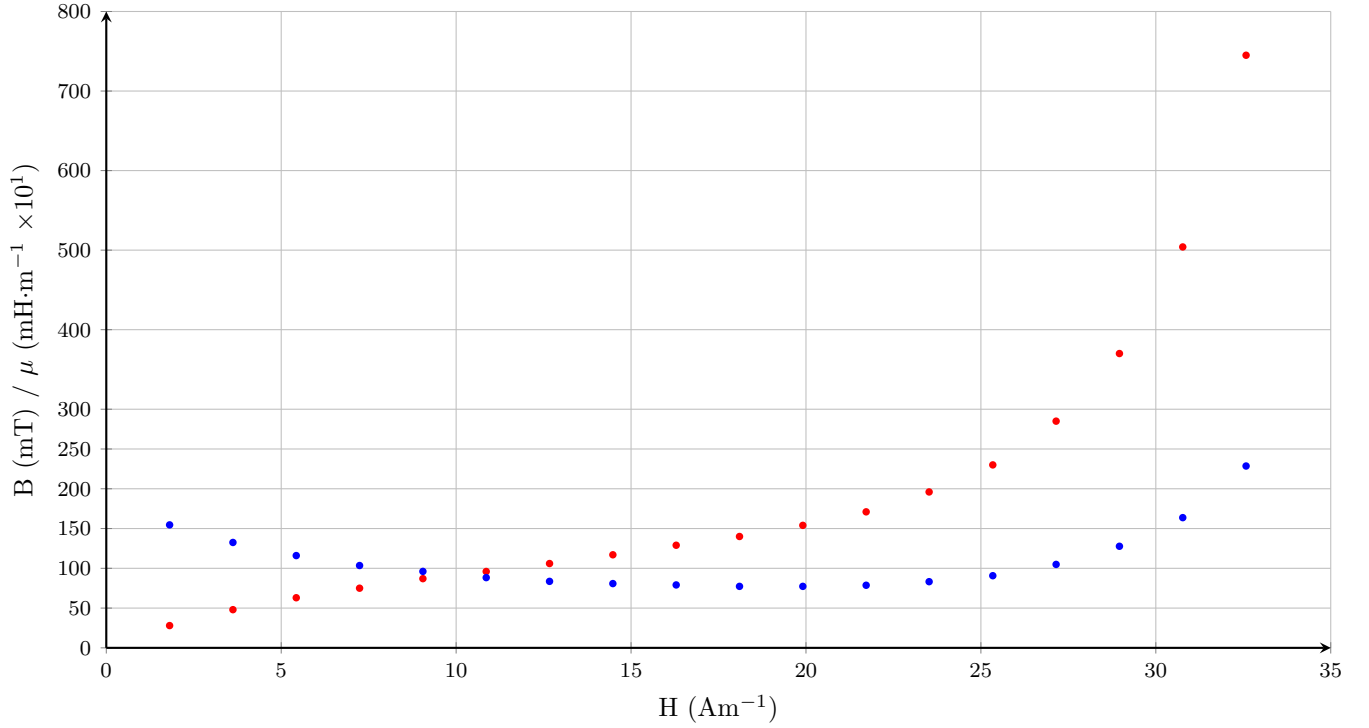


Figure 5: Sample A's relationship between magnetisation B (red) and permeability μ (blue) with external field strength H . Permeability has been scaled by a factor of 10.

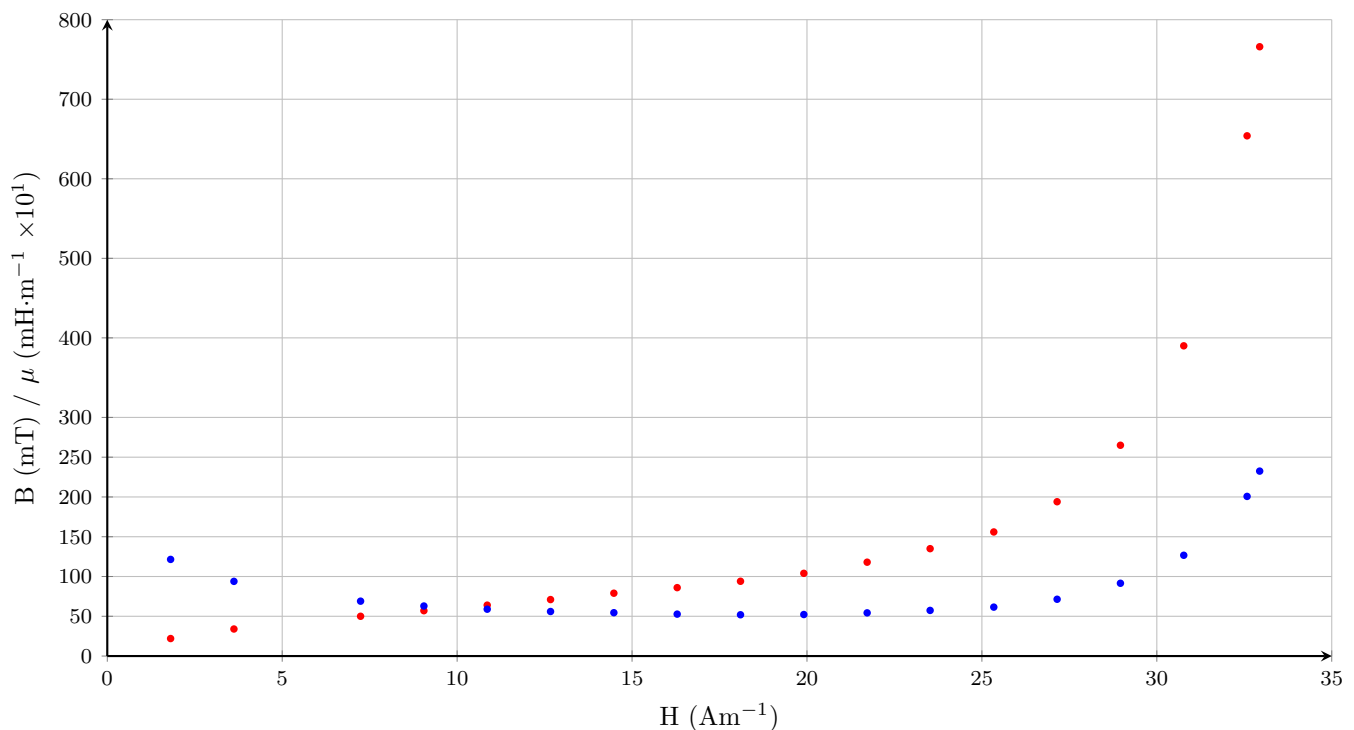


Figure 6: Sample B's relationship between magnetisation B (red) and permeability μ (blue) with external field strength H . Permeability has been scaled by a factor of 10.

2.4 Method for Curie Temperature Measurement

1. Sample A was connected to the LEAI-71 according to Figure 10 of the lab manual.

3 Discussion

Error analysis

Discussion of the final results.

References

- [1] Wikipedia contributors. Ferromagnetism — Wikipedia, the free encyclopedia, 2026. [Online; accessed 2-May-2026].
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- [4] Department of Physics. Hysteresis. Technical report, The University of Western Australia, n.d.
- [5] Electrical4U. Hysteresis loop: What is it (and what is its significance)?, 2024. [Online; accessed 2-May-2026].